Netflix Data Analysis

SQL Project Report & Business Insights

1. Project Overview

This project uses SQL in MySQL to explore a Netflix titles dataset containing information such as title, content type, director, country, release year, rating, duration, cast, genre and date added. The purpose is to understand the composition of the content library, explore release and addition trends, and practise data-quality checks and analytical SQL techniques.

2. Tools and SQL Concepts

Tool: MySQL / MySQL Workbench
SQL concepts used: SELECT, WHERE, GROUP BY, ORDER BY, aggregate functions, CASE, subqueries, CAST, SUBSTRING_INDEX, STR_TO_DATE, JSON_TABLE, and the RANK() window function.

3. Analysis Performed

- Dataset structure and sample records
- Total record count and content-type distribution
- Missing-value checks and duplicate show_id checks
- Most common content rating
- Top directors and frequently occurring actors
- Movie releases by year and content released before 2010
- TV Show seasons and movie duration
- Genre and country distribution
- Average release year by content type
- Percentage distribution of Movies versus TV Shows
- Country ranking using RANK()
- Comparison of release year and year added to Netflix

4. Business Insights

Content mix: The Movies-versus-TV-Shows breakdown shows how the catalogue is split between the two content types. This can help describe the platform's content mix and support planning for different viewing formats.

Content maturity and audience suitability: The most common rating and the distribution of ratings help summarise the audience categories represented in the catalogue. These results can support content categorisation and audience-focused reporting.

Content acquisition and release timing: Comparing release year with the year added to Netflix helps distinguish recently released titles from older titles added later. This can inform analysis of catalogue refresh and content acquisition patterns.

Country representation: Country counts and rankings help identify which country labels appear most often in the dataset. This can support a high-level view of geographic representation, with care taken when a row lists multiple countries.

Genre and creative contributors: Genre, director and actor analyses help identify frequently represented categories and contributors. These patterns can be used as descriptive context for catalogue planning, but do not by themselves prove popularity or viewership.

Historical catalogue trends: Year-wise movie counts and pre-2010 content counts show how the catalogue spans release periods. This helps describe the balance between older and newer releases.

5. Data Quality and Query Review Notes

- **Important:** The uploaded SQL file contains queries, but not the result tables/output values. Therefore, this report does not invent totals, percentages, top countries, or top genres. Add the actual MySQL outputs if you want numeric findings.
- The first TV-show seasons query uses an empty string as the SUBSTRING_INDEX delimiter, which should be corrected to a space. A later version in the file already uses a space delimiter.
- The query titled “Movies released in the last 5 years” currently sorts movies by release year and returns the latest 10 rows; it does not filter to a five-year window.
- The genre query groups by the full listed_in value. If a row contains multiple genres, it counts the combined genre string, not each individual genre separately.
- The country queries group by the full country field. If a row contains multiple countries, that combined value is treated as one label.
- The duplicate check identifies show_id values appearing more than once, but its output must be reviewed to confirm whether duplicates exist.
- The missing-value check covers selected columns only; it is not a check of every column in the dataset.

6. Conclusion

This project demonstrates the use of SQL to inspect a dataset, check selected data-quality issues, summarise catalogue characteristics, and apply analytical techniques such as subqueries and window functions. The next step is to attach the actual query outputs and use them to state evidence-based findings with exact numbers.

Prepared from the supplied Netflix SQL script. Numerical findings are intentionally not stated because query outputs were not included in the file.